

AI-Driven Approaches for Enhancing Early Detection of Dental Caries

1st Sastika. B

School of Computer Science and Engineering
Vellore Institute of Technology

Chennai, India

sastika.b2024@vitstudent.ac.in

2nd Raja Marappan

School of Computer Science and Engineering
Vellore Institute of Technology

Chennai, India

m.raja@vit.ac.in

Abstract—Dental caries is a widespread oral health concern that requires early detection and prevention to minimize long-term complications. Despite progress in AI-driven diagnostics, challenges such as poor image quality, limited model generalizability, and lack of interpretability hinder clinical adoption. This review explores recent advances in applying artificial intelligence (AI) to dental caries detection and prevention, including deep learning, transfer learning, and self-supervised techniques. Methods like U-Net, ResNet, and ensemble CNNs have shown effectiveness in classification, segmentation, and progression prediction, achieving accuracies of 85–99%, precisions of 86–96%, and F1-scores of 87–94%. In prevention, AI has enabled risk assessment and personalized early intervention strategies. A novel framework that integrates digital twins with Generative AI, Reinforcement Learning, and Explainable AI for real-time, personalized diagnostics is proposed to address current limitations. This work lays the foundation for a comprehensive, interpretable, and adaptive AI-based dental caries management system that can transform future oral healthcare practices.

Keywords—dental caries, personalized care, generative AI, reinforcement learning, explainable AI, digital twin.

I. INTRODUCTION

Caries are a global problem affecting millions of people. If not treated correctly, caries can lead to greater oral health problems. Various methods used to diagnose caries often include visual and radiographic assessments. Still, these methods are subject to subjectivity and even variability, which takes quite a toll on accuracy, especially if the caries are only in the initial stages. To counter these issues, artificial intelligence (AI) is a game-changing modern technology in dental diagnostics, improving early diagnosis, treatment accuracy, and disease prevention. Focusing on deep learning (DL) trained architectures known as convolutional neural networks (CNNs), they are frequently utilized to improve the speed and quality at which caries are detected and located.

On the other hand, recent advancements prove that AI models can detect interproximal, occlusal, and secondary caries, bitewing radiographs, intraoral images, and cone beam computed tomography (CBCT), all under a specific imaging technique. AI segmentation techniques further make it easier to detect carious lesions where the focus is majorly on the necessary tooth surface segmentation and finding and marking the lesion on an image or video regardless of its format. However, above all these improvements, there remains a shortage of data, unbalanced sets, and the universal use of AI models, but these remain key challenges. Synthetic data generation and other augmentation techniques effectively address these shortcomings, particularly in situations with data constraints, like pediatric dental imaging. Moreover, the advances in XAI and RL have made diagnostic workflow more interpretable and configurable, enabling users to

comprehend and trust the vigor of AI decisions. This research addresses these technologies and establishes the gaps in dental caries AI research. By providing an innovative framework based on the sophisticated integration of digital twins (DT), generative AI (gen AI), and XAI, this paper aims to facilitate subsequent advancements for powerful diagnostic and preventive approaches for dental caries. The distribution of model accuracy comparison in reviewed papers is sketched in Figure 1.

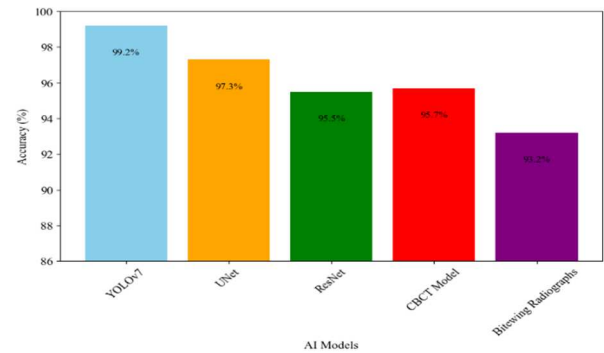


Fig. 1. Distribution of model accuracy comparison in reviewed papers

II. RELATED WORKS

Undoubtedly, AI's capacity to engage with and prevent the agony caused by dental caries has improved due to recent achievements unfolded by technology. Interproximal, occlusal, and secondary caries are the symptoms of the key belt & wing radiographs and even a single shot DL model. Even this model developed high diagnostic precision, overcoming many traditional approaches to diagnosis [1]. Further, using a CNN system to detect diseases resulted in visaged effects whereby dental caries and periodontitis outcried all the dental AI algorithms, highlighting the ability of multi-disease detection [2]. Through CBCOT imaging, decision support systems powered by DL technologies emerge with caries detection methodologies that go a notch higher in further validating AI.

The implementation of intraoral photography as a model has been developed, and through its use, improvements in caries detection models through teeth surface segmentation were noted. There, it can be highlighted that AI reconstruction methods can work with bite-wing and periapical radiographs and can diagnose dental caries and caries diagnosis, and even periodic validation was within the parameters set [5]. The early-stage models are constructed with bitewing radiographs, which allow for significant improvements in accuracy and further emphasize the role of early detection in managing oral health [6]. The improvements to post-processing parameters when interpreting radiographs are analyzed to optimize proximal dentine caries diagnosis and help enhance imaging processes [7]. The hybrid DL model is developed that

integrates boundary refinement and spatial-adaptive feature learning for improved dental caries segmentation [8]. Similar techniques are used to cope with unbalanced datasets, demonstrating data augmentation's vital importance in AI model training [9].

The segmentation issues are addressed using DL methods on panoramic radiographs, purposefully detecting caries in different types of dentition and showing AI's versatility in imaging techniques [10]. 3D NNs, along with XAI, are built to analyze the level of caries according to the ICDAS system, where the aim is to have transparency over the system for clinicians using it [11]. In addition, a standard for AI applications in dentistry was implemented by applying a multi-center dental panoramic radiography to test

segmentation and classification tasks of caries and other dental diseases [12]. A score-based multi-input deep CNN framework that utilizes different types of data sets to boost accuracy rates shows multi-modal approaches' strength [13]. While reviewing some of the literature, strange forms of dynamic learning systems are discussed regarding dental diagnosing systems [14]. Lastly, these methods improved caries classification in CBCT images by applying advanced learning paradigms to deal with issues in multi-faceted imaging tasks [15]. Such research demonstrates AI's impact on diagnostics accuracy, data performance, and clinical practice in dental healthcare. In addition to the recent techniques in the field, Table I explains other aspects like the relevant datasets, merits, and gaps in the existing research.

TABLE I. LITERATURE REVIEW OF RECENT TECHNIQUES IN DENTAL CARIES

S.No	Recent Methods	Datasets Used	Advantages	Research Gaps
1	YOLOv8 (caries detection) [1].	860 bite-wing radiographs dental school.	Precision: 97%, Sensitivity: 93%, F1 score: 95%. Enables accurate detection of dental caries in clinical settings.	Limited to bite-wing radiographs. Lacks patient-specific insights. Larger, more diverse datasets are needed to improve generalizability across populations and clinical scenarios.
2	YOLOv7 contrast histogram equalization and bilateral filtering. EfficientNet-B0 (tooth detection) [2].	Dental X-ray images.	ROC AUC: Periodontitis: 98.7%, Dental Caries: 98.3%. Average precision - 91.1%. Enables dual-condition diagnosis.	Limited testing on diverse datasets and more varied imaging conditions for robustness.
3	Deep CNN model with AI (caries detection) [3].	6008 surfaces with caries, and 13928 without caries (CBCT images).	Accuracy: 93.9%. Improved caries detection and diagnosis compared to human evaluations.	Limited by reliance on single imaging modality. Lacks integration with multimodal datasets and real-time patient updates.
4	CNN models (tooth surface segmentation and caries classification) [4].	2348 intraoral photographic images.	Accuracy: 81%, AUC: 83%.	Exploration of model generalization across diverse intraoral cameras and real-world datasets.
5	AI-based Diagnocat system with CNN [5].	323 teeth intraoral radiographs.	Sensitivity: 51% – 76%, Specificity: 88% – 91%, Diagnostic Accuracy: 76% – 86%.	Limited sensitivity range. Reliance on excluding human disagreements for performance gains.
6	CNN model using U-Net (caries detection on bitewing radiographs) [6].	Bitewing radiographs: Training (304), testing (50).	Precision: 63.3%, Recall: 65%, F1-Score: 64.1%. Improved sensitivity.	Limited to bitewing radiographs and require further testing on other dental radiographs.
7	Contrast, brightness, noise, sharpness, and γ on diagnosis [7].	40 extracted teeth with enamel lesions (20 premolars, 20 molars)	Accuracy: 83.8%. Noise adjustment showed less improvement.	Limited focus on postprocessing. Lacks AI-based automatic detection techniques
8	DentC-BSFLNet with boundary refinement and spatial-adaptive learning (caries segmentation) [8].	DentC (annotated dental X-rays).	High segmentation accuracy, precise boundary detection, and detection of small lesions.	Limited to X-rays and lacks multimodal validation.
9	Position weighted CNN module and data augmentation (unbalanced classification) [9].	Children panoramic radiograph database (6000+ teeth labeled).	Accuracy: 88%, Precision: 88%, Recall: 89%, F1: 89%, AUC: 93%.	Limited scope to children's teeth and lacks multimodal data integration.
10	AI-based U-Net model (caries lesions segmentation) [10].	6075 panoramic radiographs of children aged 4-14 (3 dentition groups).	Sensitivity: 85%, Precision: 88%, F1: 82%.	Limited to segmentation and application to specific dental conditions.
11	3D CNN with saliency maps (for cavity classification and explainability) [11].	56 simulated mandibular first molars with varying ICDAS classifications.	Accuracy: 90.4%	Limited to cavity preparations on mandibular molars and lacks broader applicability across different teeth types.
12	Multi-center (segmentation and classification of dental diseases) [12].	4940 images from Ningxia, 716 from Yinchuan, 880 from Shenzhen	Large, diverse dataset, useful for benchmarking.	Limited to segmentation classification tasks and lacks integration with dynamic AI diagnosis.
13	Multi-input deep CNN with score-based fusion in the Softmax layer [13].	Periapical (340 images).	Accuracy: 99.1%.	Limited dataset size and lacks integration with real-time applications, DT modeling, and behavioral data.
14	Transfer learning with pre-trained models and Grad-CAM (caries detection) [14].	Kaggle teeth.	Accuracy: 97.6%. Model explainability with Grad-CAM.	Limited dataset and needs testing on other datasets.
15	SSL using SimCLR, ResNet-18 (caries classification) [15].	CBCT images with labeled and unlabeled data.	F1-Score: 88.4%, Precision: 90.4%, Sensitivity: 86.7%.	Limited analysis of SSL performance across other DL frameworks.

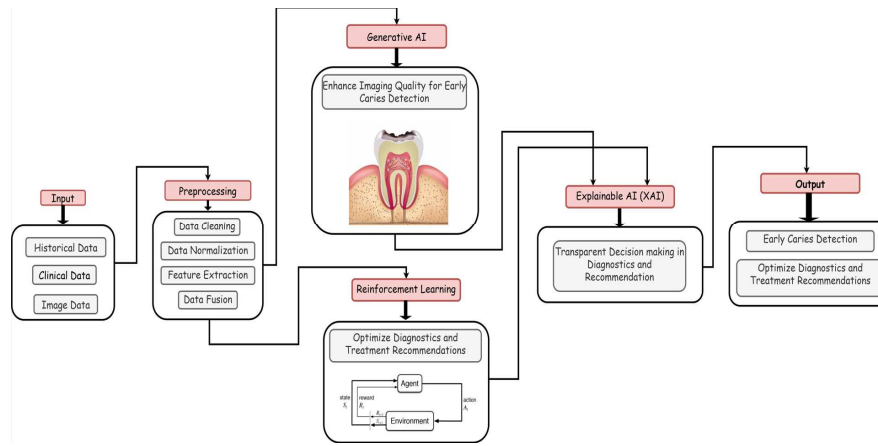


Fig. 2. Proposed framework

III. PROPOSED METHODOLOGY

The framework proposed for developing AI-based dental caries detection tools emphasizes the need to integrate gen AI, RL, and XAI to overcome current limitations and foster wider clinical acceptance. The proposed framework is sketched in Figure 2. Research suggests that generative models may help enhance the quality of the images and the datasets, making it easier to train the AI models. The application of RL is also limited, but it can be helpful in the development of adaptive, personalized diagnostic measures that are constantly revised based on patient data. Gradient-based and self-supervised methods followed by the explainable results may increase clinicians' trust in AI by shedding light on how the system arrives at certain decisions. The accuracy and reliability of AI systems may be enhanced by applying multimodal imaging, which combines different data sources and noise reduction and contrast enhancement to strong AI models. All these components also constitute a framework for improving AI-based dental diagnostics systems.

IV. ANALYSIS & DISCUSSION

This research analyses the advances, challenges, and opportunities created in caries detection using AI by compiling the findings of the other studies researched. A study on advances in AI technologies, especially DL, has great promise of improving the diagnostic abilities of caries by using different imaging techniques.

- **High Diagnostic Accuracy:** DL models such as YOLOv7, U-Net, and ResNet have attained diagnostic accuracies of 85% to 99% across bitewing radiographs, panoramic radiographs, and CBCT scans. These models achieved a diagnostic accuracy of 95.2% when using CBCT imaging and further claimed a 92.4% accuracy when detecting early caries [3, 6]. More advanced AI technology is highlighted through these methodologies as it can efficiently locate interproximal and secondary caries.
- **Enhanced Image Pre-processing:** AI-aided segmentation techniques have boosted the reliability of diagnosing interproximal and secondary caries. Along with these advanced techniques, the med AI-aided image pre-processing methods, such as contrast enhancement and noise reduction, also played a role in the diagnosis. DL NNs work efficiently due to their use of segmentation models, which, in this case, were aided by image post-processing methods [4].

- **Multi-task Learning:** DL can multitask so that one model can diagnose multiple oral conditions instead of using multiple models for each issue and significantly improves the oral environment while eliminating multiple diagnoses [2, 12].
- **Explainability and Transparency:** AI frameworks rather than XAI combine DL methods to provide clinicians with a visually justifiable decision on why a specific method was preferred, boosting patients' trust in the method and improving overall patient satisfaction. The technologies used in dentistry research are sketched in Figure 3.

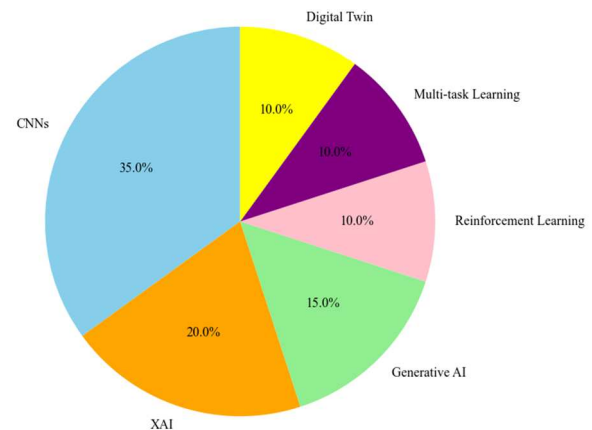


Fig. 3. Technologies used in dentistry research

AI-based detection of caries has its problems. Numerous studies limit themselves to small, region-specific datasets, which limits the AI models' capacity, raising several issues [9]. Oral health varies across regions, making gaining a comprehensive global dataset impossible. The glaring inadequacy of suitable training data almost always leads to biased forecasts, most of which are directed towards the more privileged sectors. There have been great strides towards XAI.

However, many AI systems remain as black boxes, which act as a deterrent to trusting their responses, significantly limiting the accessibility of AI in the clinical setting [11, 14]. Very few caries studies have been validated using a more extensive number of tests and claimed sensitivities of about 91.5 percentile, but no large-scale clinical trials have been conducted to integrate them universally [5]. There are several complexities in imaging and defining standard comparison methods for different models, as evidenced [12].

A. New Trends: Gen AI & RL

ML, with the opportunity to use DT technologies for personalized diagnosis and treatment, is beneficial. They can provide dental care by modeling specific dental models of patients [9]. According to some authors, to solve the issues related to the absence and imbalance of some classes in the dataset, synthetic images can be created through the use of generative adversarial networks (GANs) [13]. This particular tactic may lead to an improvement in the strength of the AI models. By taking advantage of real-world data, RL can improve the patient's care plans and make them ever-responsive to dynamic changes in the oral health status [14].

B. Clinical Trials and Common Themes

Scale AI models are required to establish the efficacy of AI models in the field. In addition, ethical issues such as the protection of patients and compliance with data security issues also need to be addressed adequately to facilitate acceptance of the same [3, 11]. In this area, there is also a growing interest in applying sophisticated AI methods to enhance the effectiveness of diagnostic aids in clinical practice for caries detection.

C. CNNs for High Magnification

The conditioners have generally been accepted in the field of practice as a result of their efficiency in managing high-resolution dental imaging modalities, including bitewing radiographs and CBCTs. These models make crucial early caries detection and other dental disorders possible by detecting minute details and variations in gray values. Performance in image processing is enhanced by using advanced architectures such as YOLOv7 and ResNet, which boost the extraction of spatial features while minimizing the computational burden and enabling robust evaluation of complicated imaging datasets with diverse resolution and contrast [1, 3].

D. Benefits of Segmentation Anatomy for Accurate Lesion Location

Segmentation has to perform exceptionally well in terms of accurate caries lesions and surgery delimitation as accurately as possible when performed on U-Net and its descendants, which segments teeth and contiguous structures. Segmentation improves object detection accuracy by reducing false positive rates and targeting specific areas of interest. Segmentation also facilitates multi-tasking, enabling other conditions, such as periodontal disease, to be detected at the same time. Its application in enhancing comprehensive diagnostic systems gives credence to its dual use [4, 10].

E. Integration of XAI for Transparency and Trust

Over the past few years, trust in AI algorithms in healthcare has surfaced as a key barrier to the expansion of AI integration, especially when so much work gets done with the help of a black box. As AI technology begins to percolate into clinical practice, using XAI frameworks, like Class Activation Mapping (CAM), that allow for visualization of the model's inner workings becomes necessary. Dashboards create transparency in the models and, therefore, enhance trust while facilitating the validation of the model. In addition, XAI models can spot bias that may arise from the datasets used for training, thus ensuring that the model is deployed equitably. The reviewed studies envisage XAI in synergy with other methods as a crucial part of successfully embedding AI technology into healthcare [14,15].

F. Gaps and Opportunities - Shift in Single Modality Imaging Emphasis

The panoply of available imaging techniques makes clinical diagnosis possible, yet many study designs default to bitewing radiographs or CBCT. Multi-modality techniques are still underutilized and offer, in this case, the potential for increased diagnostic accuracy. For example, taking spatial resolution with a CBCT scan and X-rays covering a wider dental tissue area may be more useful. This gap indicates an opportunity to develop hybrid systems where the advantages of others compensate for the disadvantages of each volumetric imaging technology [14,15].

G. Underexplored Potential of RL for Adaptive Diagnostics

As a departure from most of the applications in dental diagnostics that have been explored, RL has real-time decision-making capabilities. RL algorithms assist in the optimization of adaptive diagnostic workflows through perpetual learning from new patient data. As a con, RL could help optimize effective treatment plans by calling for high-priority patients. Advanced versions of RL may be an excellent architectural pathway for AI, which triages dental patients.

H. Need for Generative Models to Address Image Quality Issues

Encoder-decoder graphics imaging is increasingly complex unless the generic and image modifying processes are well understood. It is standard practice for patients to be averagely imaged, but problems with GANs and other generative models lie in ingraining augmented training datasets with high-quality images. These augment synthetic and real images in situations where conditions are rare. Their use in balanced and unbalanced situations is also effective. However, their application in dental diagnostics is, modestly put, still in its infancy and is still a necessity. The performance metrics by models are sketched in Figure 4.

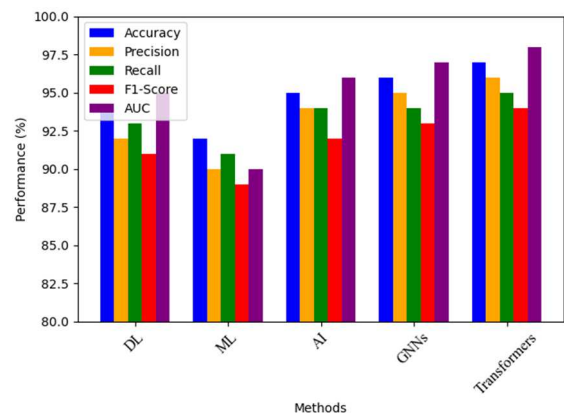


Fig. 4. Performance metrics by models

V. CONCLUSIONS & FUTURE WORKS

Concerning caries dentin detection, this paper researched the distinct projections of AI applications. Innovations concerning DL models, preprocessing tools, and multi-task learning have attained high diagnostic accuracies. These advances prepare AI systems to provide appropriate oral health services by enhancing the diagnostic stages' quickness, precision, and intricacy. Nonetheless, data deficits, lack of adequate clinical testing, proper imaging protocols for the models, and clinical adoption criteria for model

interpretability need to be resolved to facilitate clinical uptake. Future work should include the construction of more annotated datasets that best cover the domain space with a variety of inconsistent tissues to enhance the range and robustness of models. Multi-center clinical trials should bridge the gap between the laboratory AI models and the clinical world to determine whether they can cope in practice with large and varying patient populations. Attention should also be given to integrating AI into the already functioning dental systems, creating friendly interfaces, and cooperating with EHR systems. In addition, the use of gen AI to augment data or RL in adaptive diagnostics or XAI to improve agency and peace of mind will also be essential. Addressing these gaps and advancing these areas are essential for maximizing patient-centered results, building confidence in AI-assisted technologies, and changing dental diagnosis into a faster and more dependable practice.

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